

Vision Centric Tic Tac Toe Robot on Dobot Magician Lite Using Minimax and OpenCV

Jatin Satyam

School for Engineering of Matter, Transport and Energy, Ira Fulton Schools of Engineering
Arizona State University, Tempe, Az
Email: jsatyam@asu.edu

Abstract—This report describes the development of a semi-autonomous Tic-Tac-Toe robot built on the Dobot Magician Lite. The system uses an overhead camera as its only sensor and relies on classical computer-vision techniques to locate the game grid and interpret human moves. The robot draws the board, recognizes X/O marks through edge detection and geometric cues, validates each move, and responds using a Minimax policy. The robot writes all outcomes directly on paper, making the perception-action loop easy to observe. Although the full pipeline is still under stabilization, core components—motion control, grid drawing, camera calibration, and symbol classification—have been successfully demonstrated. The remaining challenges mainly involve improving the scan loop reliability and creating more robust safeguards against frame dropouts and lighting variations.

Index Terms—Perception-action, Forward Kinematics, Inverse Kinematics, Homogeneous Transformation Matrices, Camera Calibration.

I. INTRODUCTION

Tic-Tac-Toe is a simple game computationally, but converting it into a full robotic demonstration exposes all essential components of autonomy: perception, decision-making, control, and human interaction. Instead of using a simulated environment, this project embraces the real world—paper shifts, camera noise, uneven lighting, and imperfect drawing. The robot draws its own grid, detects it visually, and then uses these detected lines to define region-of-interest boundaries for symbol classification. No manual calibration is needed. Everything is self-aligned with what the robot physically produces.

To interact with the human player, the robot waits for visible changes on the page rather than any digital inputs. Only one new symbol is accepted per human turn; any inconsistencies lead to an “E” mark written on the paper. This keeps every rule transparent and physically visible.

II. THEORY

Prior robotic board-game systems commonly rely on fiducial markers, templates, or controlled lighting environments to simplify perception. While effective, such methods impose setup overhead and reduce portability. Recent classroom-friendly approaches favor minimal calibration and geometry-based perception, similar to the strategy used here.

Optimal decision-making for Tic-Tac-Toe is well established in literature, with Minimax providing guaranteed optimal play

in a small search space. Unlike reinforcement-learning approaches, Minimax is deterministic, explainable, and requires no training data.

Robotic drawing has been explored in plotters, artistic robots, and calligraphy systems. Our implementation adapts these ideas with emphasis on legibility and reliability rather than aesthetic precision.

III. SYSTEM ARCHITECTURE

A. Hardware Setup

- Dobot Magician Lite robotic arm.
- Pencil holder with writing depth 0.2 mm.
- Overhead UVC camera positioned 50 cm above the page.
- A4 sheet taped to prevent movement.
- Typical grid cell size: 25–32 mm.

B. Software and Framework

- Ubuntu + Python 3
- OpenCV for perception
- pydobot for robot motion control
- NumPy for geometric operations
- V4L2 for camera initialization

IV. METHODOLOGY

A. Grid Drawing and Calibration

The robot draws two horizontal and two vertical lines using predefined physical coordinates. After drawing, the system captures frames and applies:

- Grayscale conversion
- Noise reduction
- Canny edge detection
- Probabilistic Hough Transform

Detected lines establish the board quadrilateral, and a homography rectifies the grid into a square representation. Calibration completes once consistent line geometry is detected.

B. Symbol Classification

Each of the nine cells is extracted as a Region of Interest (ROI).

- Classification proceeds as follows:
- Circle detection (HoughCircles) → identifies O
- Diagonal line detection (HoughLinesP) → identifies X
- Ink-density fallback → resolves ambiguous cases

A stabilization filter requires the same classification result across N consecutive frames, reducing errors caused by flicker or lighting variation.

C. Move Validation

Let b_t and b_{t+1} denote two consecutive board states captured by the vision system. A human turn is considered valid only if the following conditions are satisfied:

- Exactly one new symbol appears on the board.
- No previously written symbol is erased or overwritten.
- The new mark corresponds to the human player's assigned symbol.

To determine the correctness of each move, the system computes the difference $\Delta b = b_{t+1} - b_t$, where positive entries correspond to newly added marks and negative entries correspond to removed or altered marks. If the system detects multiple additions, erasures, or placement of the robot's symbol by the human, the move is flagged as invalid and the robot writes an "E" beside the grid to indicate an error state.

The validation procedure ensures rule compliance without requiring any digital input from the user. All interaction relies entirely on vision-based feedback from the camera.

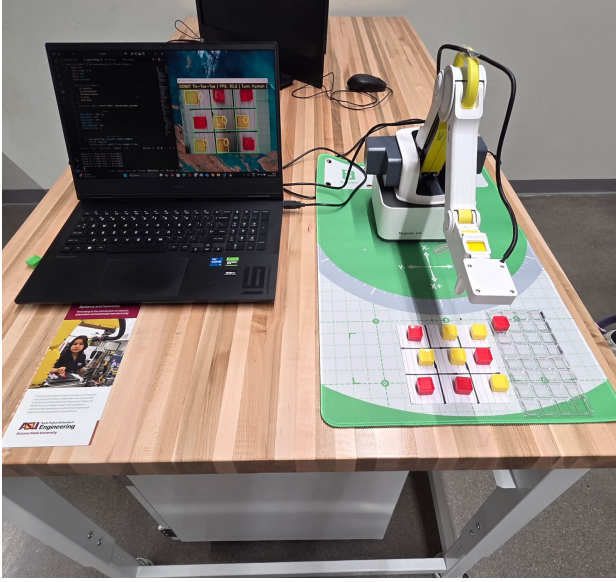


Fig. 1. Experimental setup for the Midterm-1 Tic-Tac-Toe project. The Dobot Magician Lite interacts with the physical game board while an overhead camera and a laptop run the vision pipeline, symbol classification, and Minimax control loop. The robot is shown in the process of interpreting human moves and placing game pieces on the board.

D. Decision Making (Minimax)

The robot selects its moves using a depth-aware Minimax search. Let b represent the current board configuration and d the search depth. Terminal states are scored according to:

$$\text{score}(b) = \begin{cases} +10 - d, & \text{robot wins,} \\ -10 + d, & \text{human wins,} \\ 0, & \text{draw.} \end{cases}$$

During its turn, the robot maximizes the score, while the human turn is modeled as minimizing the score. For all legal actions, the algorithm recursively explores future states until a

terminal configuration is reached. The optimal move is given by:

$$m^* = \arg \max_{m \in \mathcal{M}(b)} \text{Minimax}(b', d + 1),$$

where b' is the board resulting from applying move m .

Because Tic-Tac-Toe has a small state space, full-depth Minimax search is computationally feasible and guarantees optimal play. The depth term biases the robot toward achieving faster wins and delaying losses, resulting in more human-like but still optimal behavior.

V. RESULTS

A. Functional Components

The system successfully demonstrated reliable operation of several core modules. Grid drawing was consistent across multiple trials, with the robot producing straight, evenly spaced lines suitable for downstream detection. Camera calibration via Hough-based line extraction achieved stable quad detection whenever lighting conditions were uniform. Symbol classification for both X and O marks performed accurately under standard indoor illumination, particularly when ink contrast was high. The human move validator correctly identified illegal states such as multiple marks, overwriting, or placement of the robot's symbol. The Minimax-based decision-making module consistently produced optimal moves and demonstrated correct behavior in all tested board configurations.

B. Failure Modes

The primary obstacle to achieving fully autonomous end-to-end gameplay resulted from instability in the scanning pipeline. In several runs, the vision loop stalled during frame acquisition, causing the system to hang. CPU load occasionally spiked during symbol classification, leading to delayed responses or blocked frame reads. Reflections on the paper surface or minor camera tilt sometimes interfered with grid detection, preventing successful homography estimation. Race conditions between perception and motion callbacks were observed, particularly when the robot was drawing and the camera attempted classification simultaneously.

C. Diagnostic Observations

Log analysis suggests that the failures stem from engineering limitations rather than conceptual issues. Missing timeout conditions caused the scan loop to wait indefinitely for new frames. The vision module lacked retry mechanisms for Canny or Hough failures, leading to brittle behavior under variable lighting. The absence of frame debouncing made the classifier sensitive to transient motion or vibration. These issues collectively caused intermittent freezing but can be addressed with enhanced defensive programming.

VI. DISCUSSION

Despite the incomplete end-to-end execution, the system demonstrates that the overall architecture is sound. The robot consistently succeeded in drawing the grid, performing calibration, validating human moves, and selecting optimal responses

using Minimax. The limitations observed in the scan loop are not fundamental and can be resolved by adding watchdog timers, implementing non-blocking camera reads, improving exposure control, and refining threshold values for classification. The system remains modular, and improvements in the vision component are expected to yield a stable, fully autonomous pipeline.

VII. CONCLUSION

This midterm project implemented a vision-based Tic-Tac-Toe robot using the Dobot Magician Lite, leveraging classical computer vision techniques and a Minimax agent for optimal gameplay. Key components—including grid rendering, camera-based board calibration, symbol detection, and move validation—performed successfully in controlled conditions. The only major limitation preventing full autonomous gameplay was instability in the scanning subsystem, primarily due to timing and robustness issues in the perception pipeline. Future work will focus on adding watchdogs, improving lighting robustness, implementing graceful recovery logic, and expanding the system to support repeated gameplay on a single sheet. With these enhancements, the system is expected to achieve stable, repeatable, and interpretable performance.

ACKNOWLEDGMENT

The author would like to thank **Prof. Sangram Redkar** for his guidance throughout the RAS 545 course and laboratory sessions. Gratitude is also extended to **Sai Srinivas Tatwik Meesala** and **Rajesh S. Aouti** for their assistance during the experiment and for providing valuable insights into Dobot operation, calibration, and safety.

VIDEO PRESENTATION LINK

<https://youtu.be/YnNlS2I5-H8>

REFERENCES

- [1] R. Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed. Springer, 2022.
- [2] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, 2020.
- [3] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.
- [4] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.